

Internet of Things

Vipul Singh Negi

Department of Computer Science and Engineering

National Institute of Technology Rourkela-769008

ESP 32

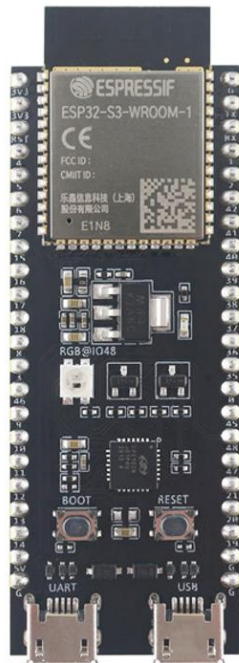
When it comes to IoT, the ESP32 is a chip that packs a powerful punch. The ESP32 is a follow-up to the ESP8266. This low-cost system on a chip (SoC) series was created by Espressif Systems. Based on its value for the price, small size, and relatively low power consumption, the ESP32 is well-suited to a number of different IoT applications.



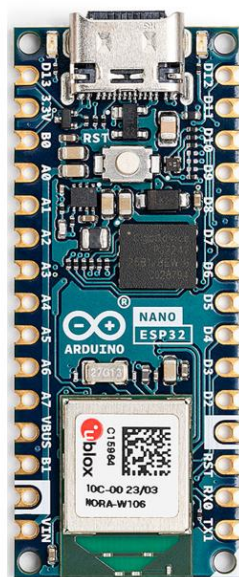
**Seed Studio
ESP32-C3**
\$4.99



**Adafruit QT Py
ESP32-C3**
\$9.95



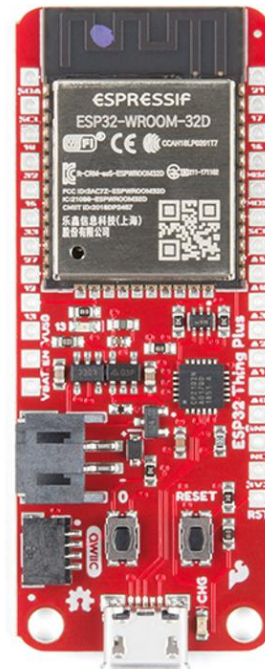
Espressif ESP32-S3 Devkit
\$15



Arduino Nano ESP32-S3
\$20



Adafruit ESP32-S2 TFT
\$24.95



Sparkfun Thing+ ESP32
\$22.50



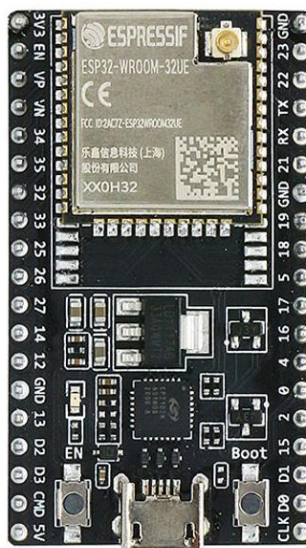
Sparkfun Thing+ ESP32-S2
\$21.50



Sparkfun Thing+ ESP32-S3
\$19.95



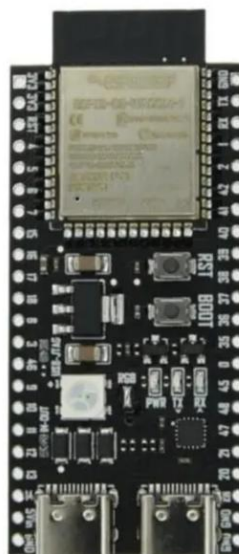
**Sparkfun Pro Micro
ESP32-C3**
\$9.95



Espressif ESP32 Devkit
\$10



Wemos Lolin ESP32-S3
\$7.09



Generic ESP32-S3
\$0.99



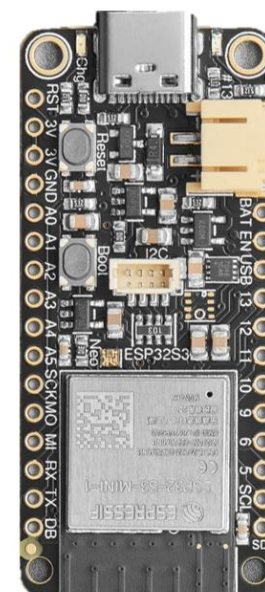
Lilygo T-Disp S3 AMOLED
\$33.98



Adafruit ESP32
\$19.95



Adafruit ESP32-S2
\$17.50



Adafruit ESP32-S3
\$17.50

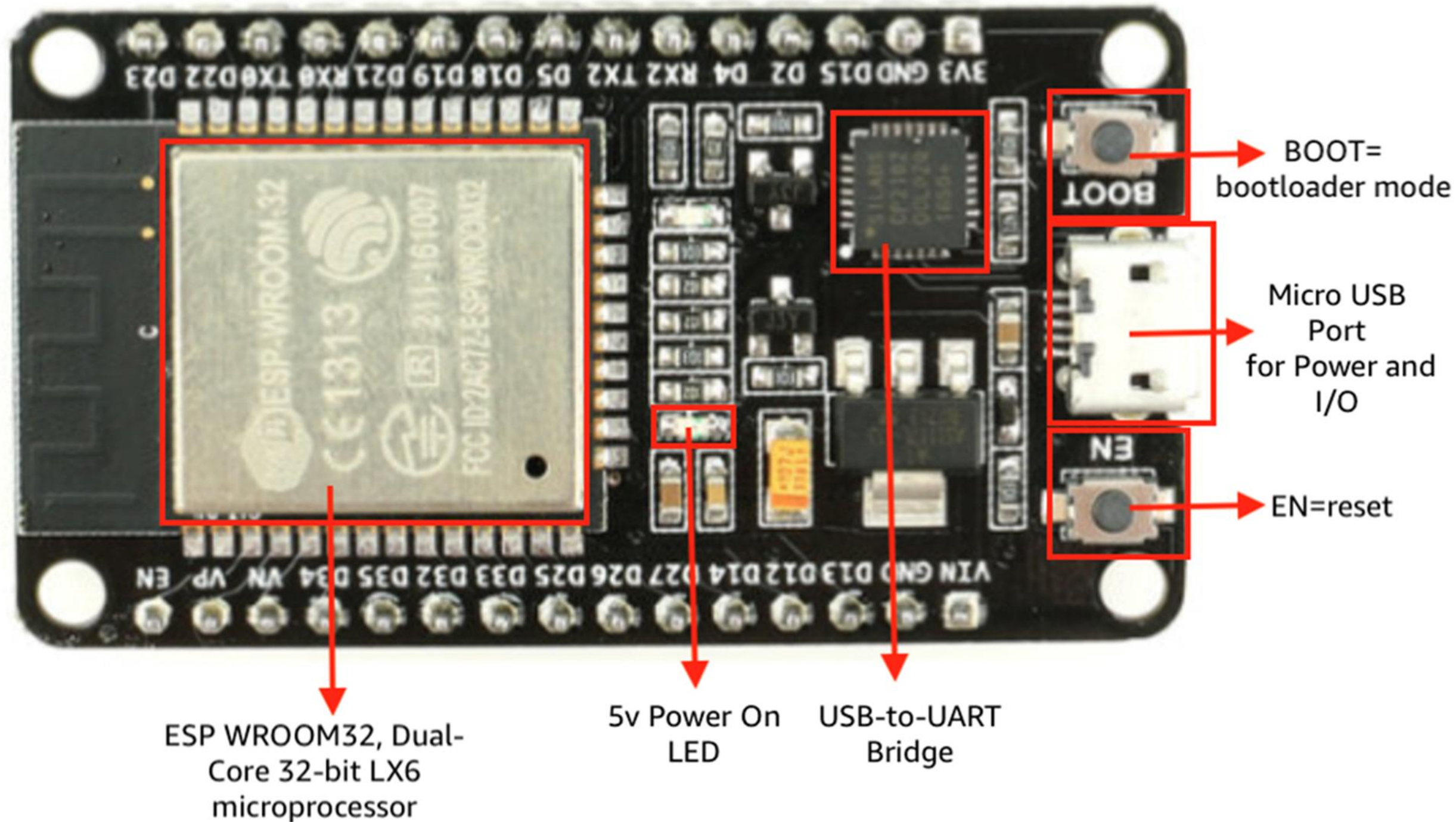
What is ESP32?

ESP32 is a series of low-cost, low-power system on a chip microcontrollers with integrated Wi-Fi and dual-mode Bluetooth.

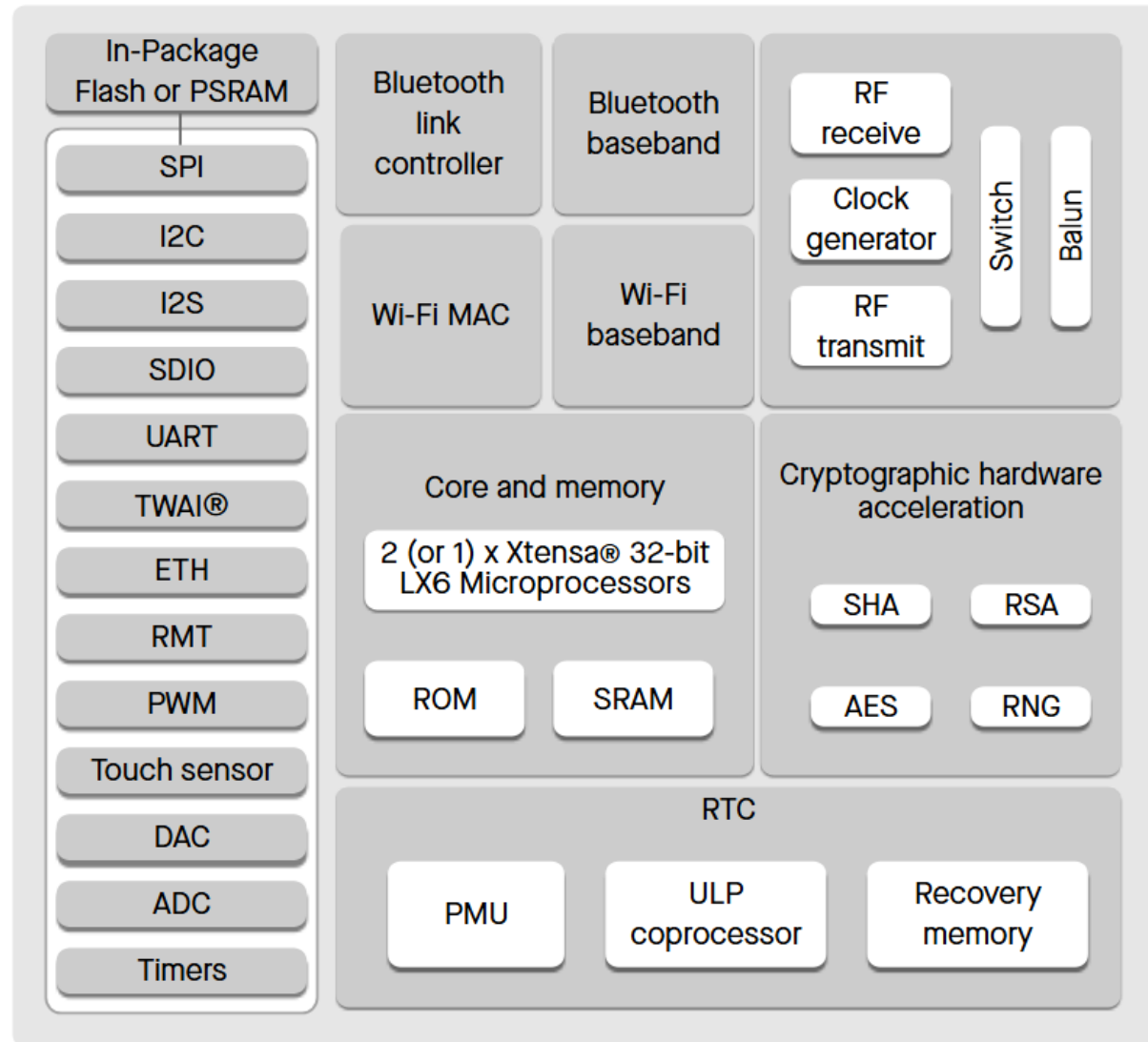
ESP32 is highly integrated with built-in antenna switches, RF balun, power amplifier, low-noise receive amplifier, filters, and power management modules. Engineered for mobile devices, wearable electronics, and IoT applications, ESP32 achieves ultra-low power consumption through power saving features

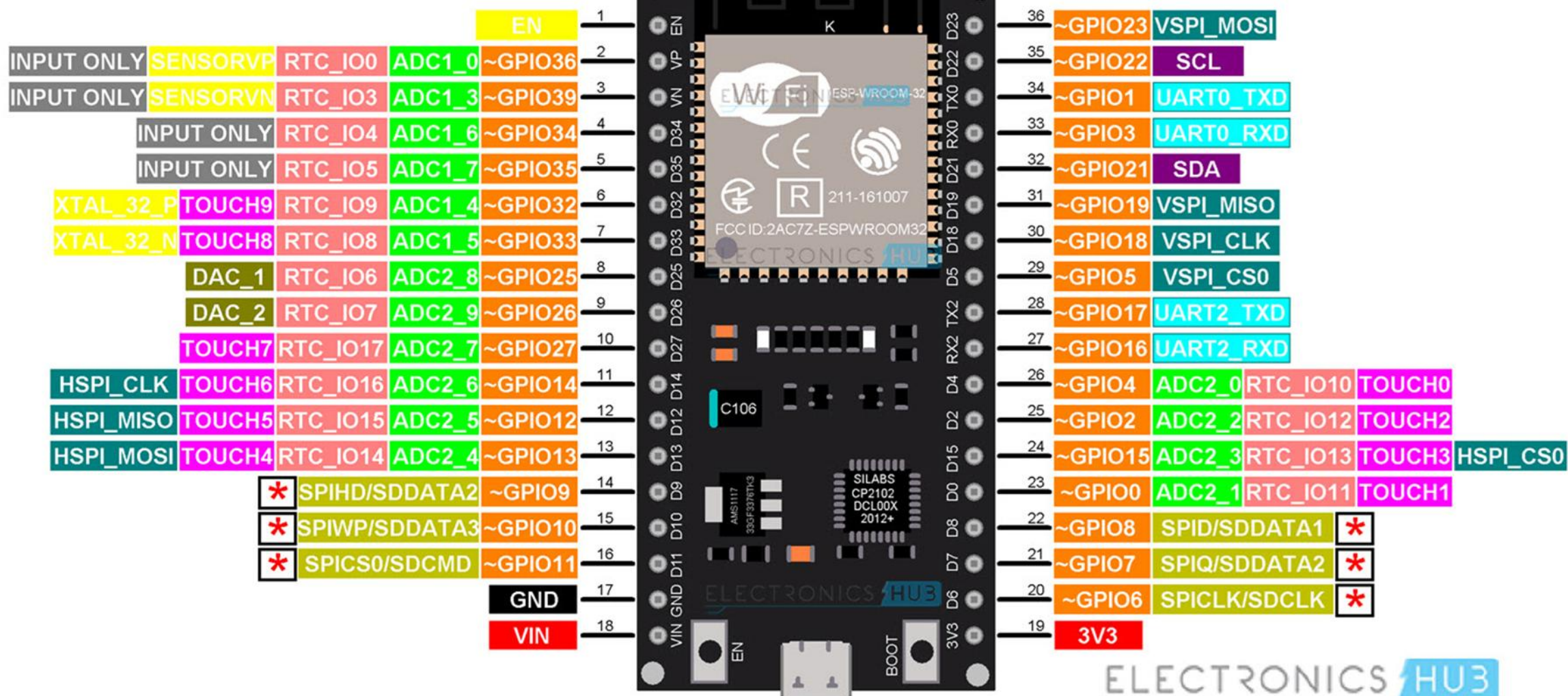
Why ESP32

- 1) It is inexpensive.
- 2) It's a Bluetooth / Wi-Fi combo module.
- 3) Extensive ecosystem support.
- 4) Options available with additional Flash and pSRAM.
- 5) Can install and run MicroPython (Give Thonny a try).



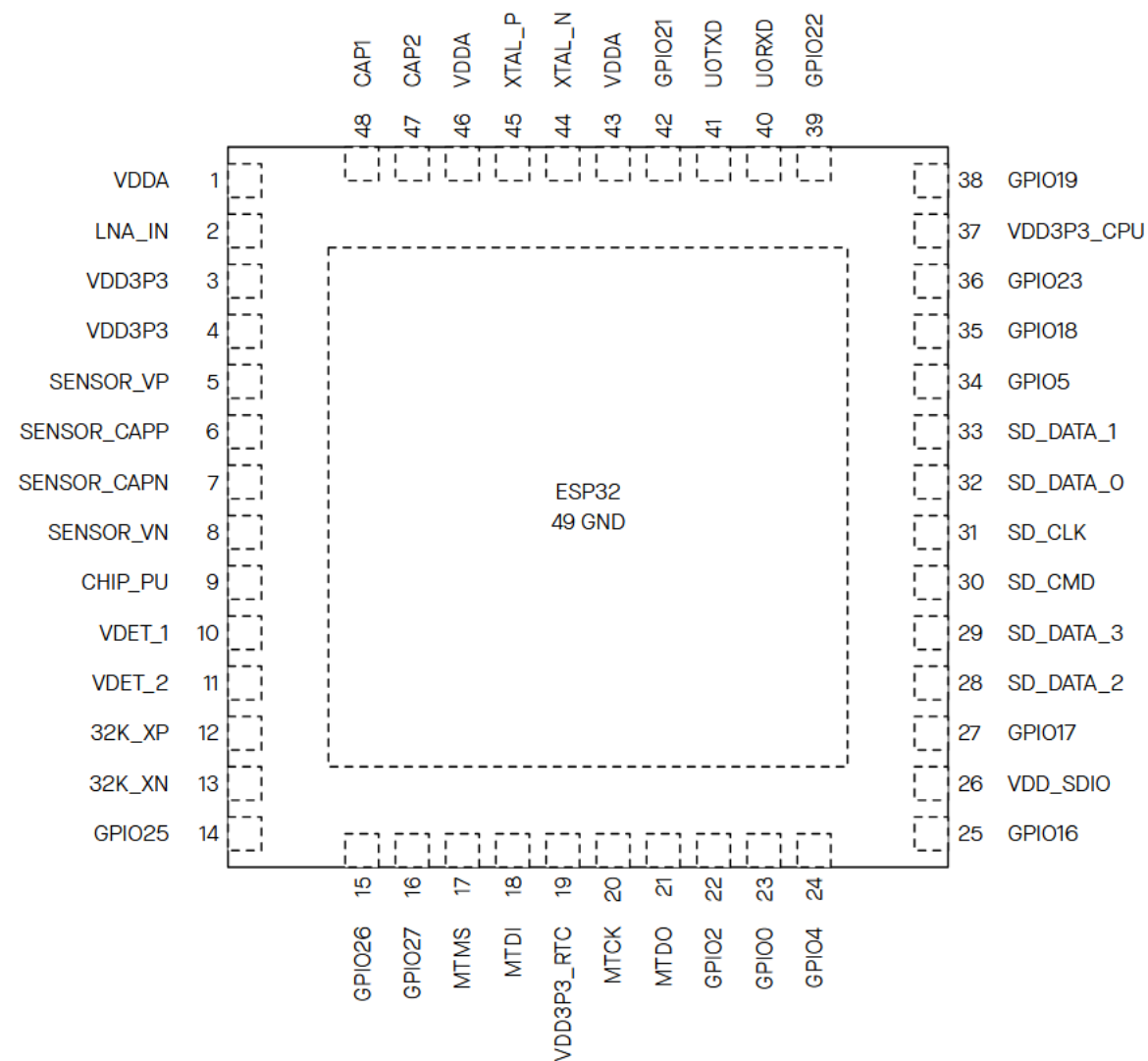
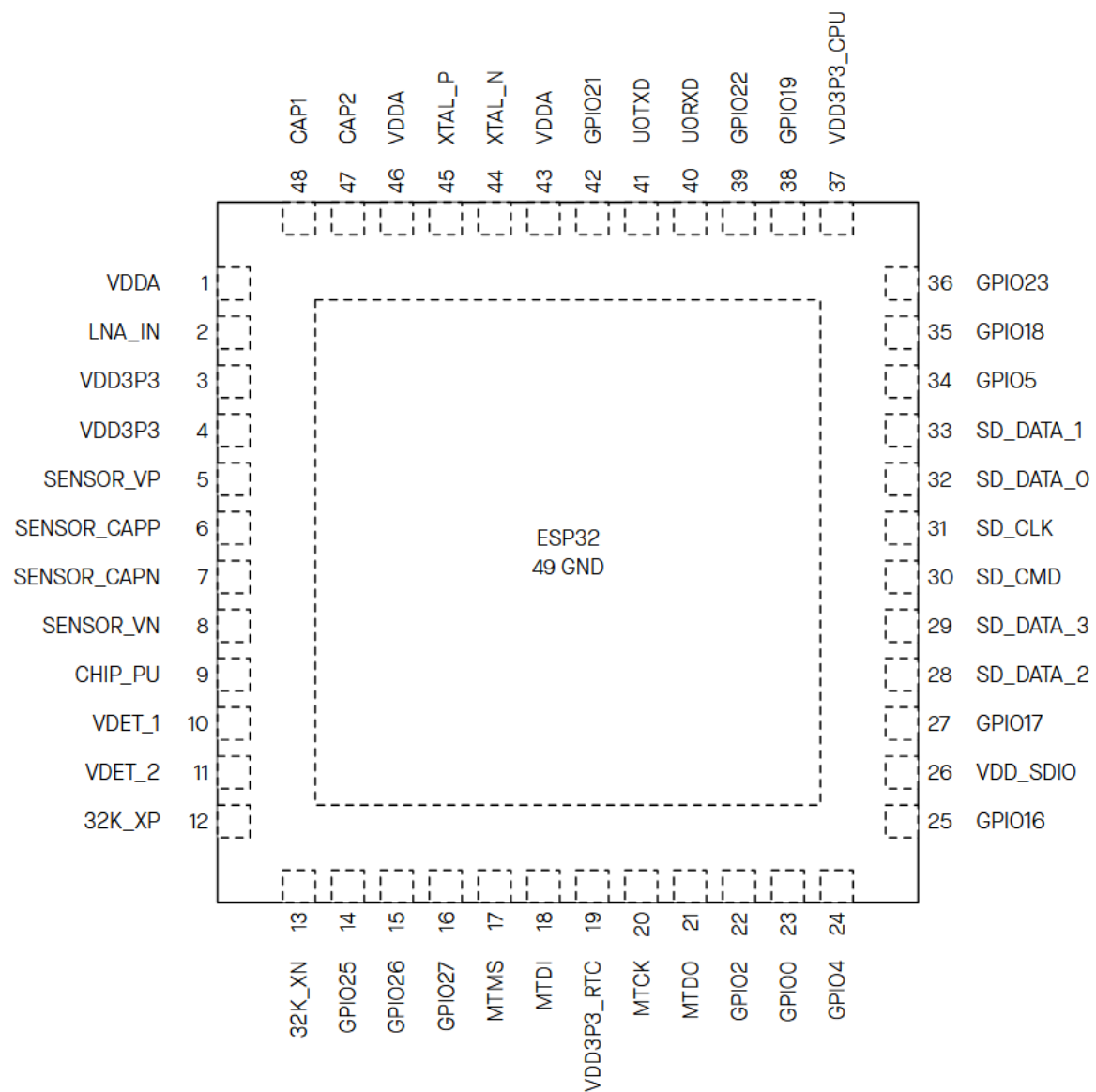
Block Diagram of ESP32

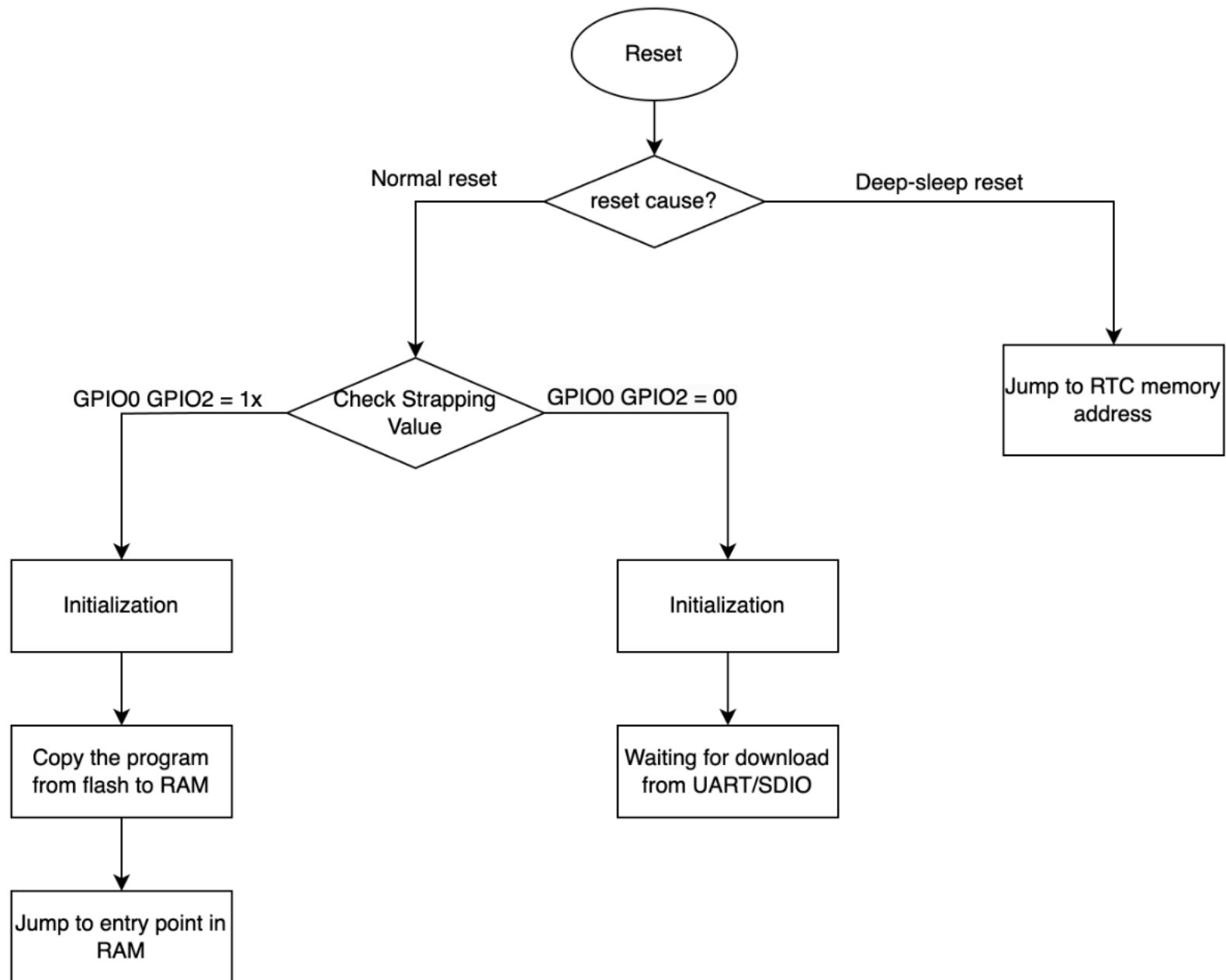




■ POWER	■ DAC	■ SYSTEM	~ PWM
■ GND	■ ANALOG	■ TOUCH	■ SPI
■ GPIO PIN	■ RTC_IO	■ I2C	

* Pins marked with this symbol are connected to SPI Flash IC. They should not be used other wise. (GPIO6 - GPIO11)





ESP-32	DESCRIPTION
Core	2
Architecture	32 bits
Clock	Tensilica Xtensa LX106 160-240MHz
WiFi	IEEE802.11 b/g/n
Bluetooth	Yes - classic & BLE
RAM	520KB
Flash	External QSPI - 16MB
GPIO	22
DAC	2
ADC	18
Interfaces	SPI-I2C-UART-I2S-CAN

Features and Specification

Processors – The ESP32 uses a Tensilica Xtensa 32-bit LX6 microprocessor. This typically relies on a dual core architecture, with the exception of one module, the ESP32-S0WD, which uses a single-core system. The clock frequency reaches up to 240MHz and it performs up to 600 DMIPS (Dhrystone millions of instructions per second). Moreover, its low power consumption allows for analog to digital conversions as well as computation and level thresholds, even while the chip is in deep sleep mode.

Wireless connectivity – The ESP32 enables connectivity to integrated Wi-Fi through the 802.11 b/g/n/e/i/. Moreover, Bluetooth connectivity is made possible with the v4.2 BR/EDR, and the series also features Bluetooth low energy (BLE).

Contd.

Memory – Internal memory for the ESP32 is as follows. ROM: 448 KB (for booting/core functions), SRAM: 520 KB (for data/instructions), RTC fast SRAM: 8 KB (for data storage/main CPU during boot from sleep mode), RTC slow SRAM: 8 KB (for co-processor access during sleep mode), and eFuse: 1 KiBit (256 bits used for the system (MAC address and chip configuration) and 768 bits reserved for customer applications). Moreover, some of the ESP32 chips, including the ESP32-D2WD and ESP32-PICO-D4, have internally connected flash.

Contd.

External flash and SRAM – ESP32 supports up to four 16 MB external QSPI flashes and SRAMs with hardware encryption based on AES to protect developers' programs and data. It accesses the external QSPI flash and SRAM through high-speed caches.

Security – The ESP32 supports all IEEE 802.11 standard security features, including WFA, WPA/WPA2 and WAPI. Moreover, ESP32 has a secure boot and flash encryption.

ESP 32 Functions

Networking: The module's Wi-Fi antenna and dual core enables embedded devices to connect to routers and transmit data.

Data processing: Includes processing basic inputs from analog and digital sensors to far more complex calculations with an RTOS or non-OS software development kit (SDK). A non-OS SDK refers to one that is designed to run directly on the chip without a full operating system supporting it.

P2P connectivity: Creates direct communication between different ESPs and other devices using IoT P2P connectivity.

Web server: Provides access to pages written in HTML or development languages.

Installation

1) File -> Preferences -> Additional Board Managers ->

"https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json" -> Ok

1) Tools -> Board -> Board Manager -> Search "esp32" -> Install
"esp32 by Espressif Systems".

2) **Download USB to UART Bridge VCP Drivers** from

"https://www.silabs.com/documents/public/software/CP210x_Windows_Drivers.zip"

1) LED using ESP32

```
1. int LED = 2;  
2. void setup() {  
3.   pinMode(LED, OUTPUT);  
4. }  
5. void loop() {  
6.   delay(50);  
7.   digitalWrite(LED,HIGH);  
8.   delay(50);  
9.   digitalWrite(LED,LOW);  
10. }
```

2)TO FIND OUT AVAILABLE BLUETOOTH DEVICES

3) TO FIND OUT AVAILABLE WIFI DEVICES

4) TO COMMUNICATE BETWEEN DEVICES USING BLUETOOTH



TO COMMUNICATE BETWEEN DEVICES USING BLUETOOTH

Download the **Serial Bluetooth Terminal** app from the playstore.

Connect your Phone to the Bluetooth of **ESP32-BT-Slave** (This name can be changed from the code).

Open the App on your phone and navigate to DEVICES and Select the **ESP32-BT-Slave**.

Assignments

1. WAP to interface the LED with the ESP32. (Create All Colors)
2. WAP to interface the LED using Bluetooth Connection with the ESP32 Development Board. (The Phone will send things like g for the color green, b for the color blue). Do this for the RGB Colors.

Thanks